ME/CE/AM 295

Agentic AI in Scientific Discovery

9 units (3-5-1)

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TAs:
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Hyoseo (Lauren) Yoon

Lecture 2

California Institute of Technology
Division of Engineering and Applied Science

Office hours

Alice, Monday after class
Lauren, Wednesday after class

Attend and discuss: Homework, Labs, Final projects and grading policies

The Dawn of Agentic AI in Engineering



From Language Models to Autonomous Agents

The Discovery Loop

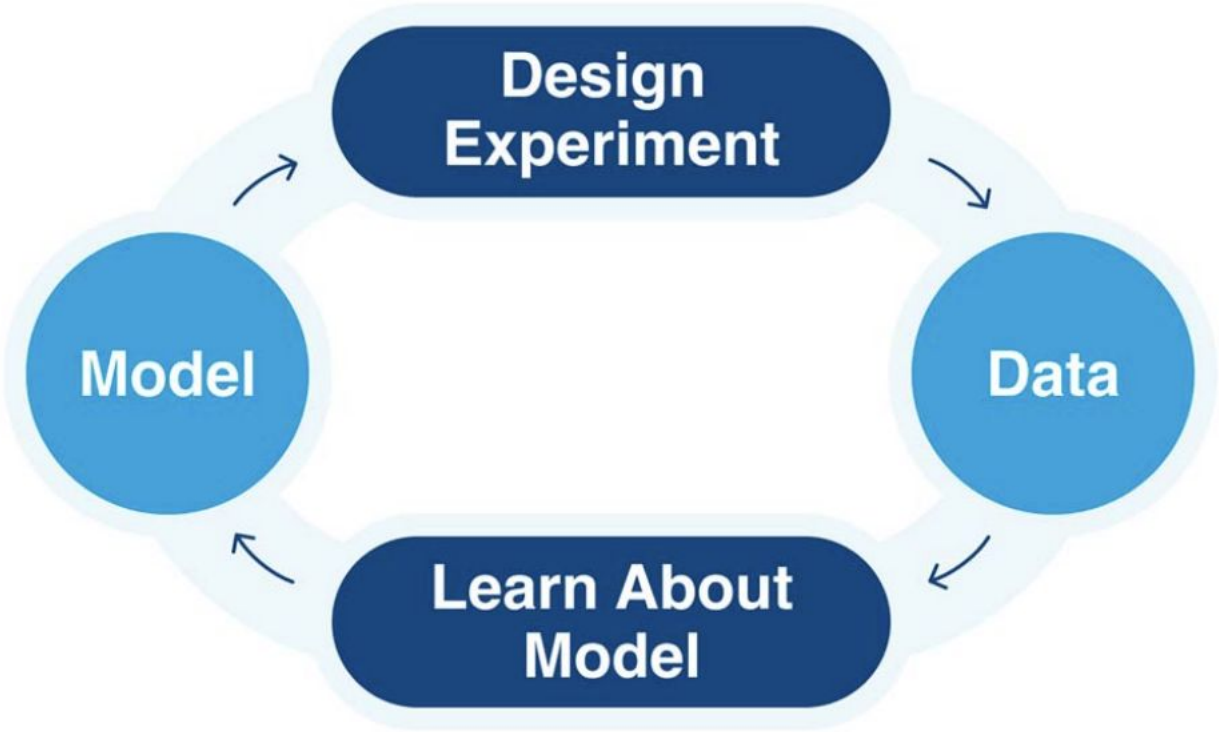
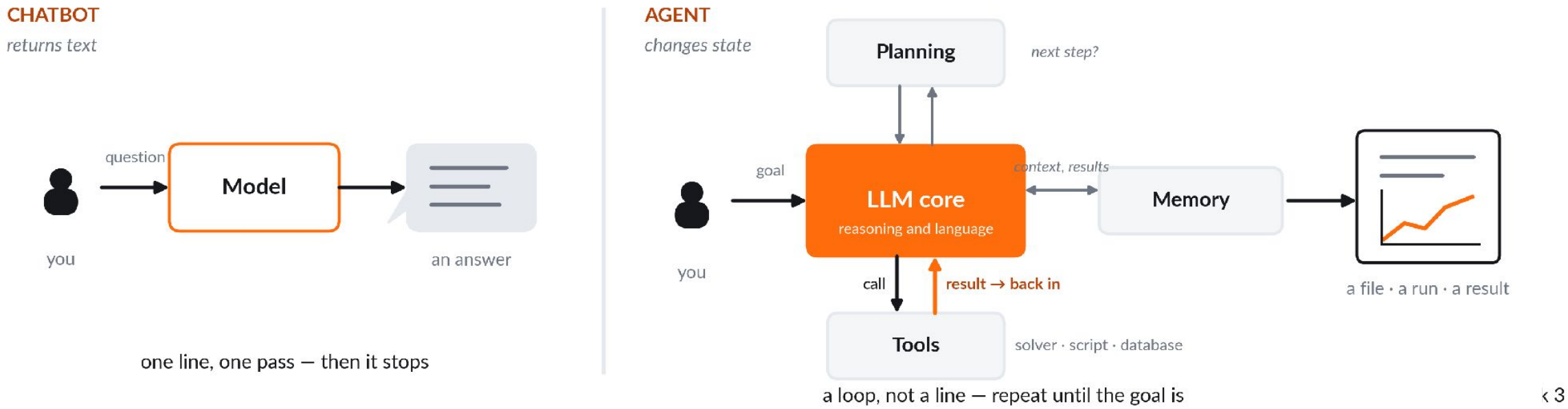


FIGURE 1. Science has progressed by iteration through a virtuous circle: models inform the design of experiments or observations; the data they produce are harnessed to inform, update, or revise models. In many fields, AI tools together with automated data acquisition now enable automation and acceleration of the entire loop, often by orders of magnitude.

Schneider et al. IEEE compiting in Science and Engineering, 2024 (10.1109/MCSE.2024.3352451)

From Answering to Doing

A chatbot answers. An agent pursues a goal: it plans, it remembers, and it uses tools.



‹ 3

LLM core
reasoning

Planning
goal → steps

Memory
carry results

Tools
solvers, data

The Agent Loop, Illustrated

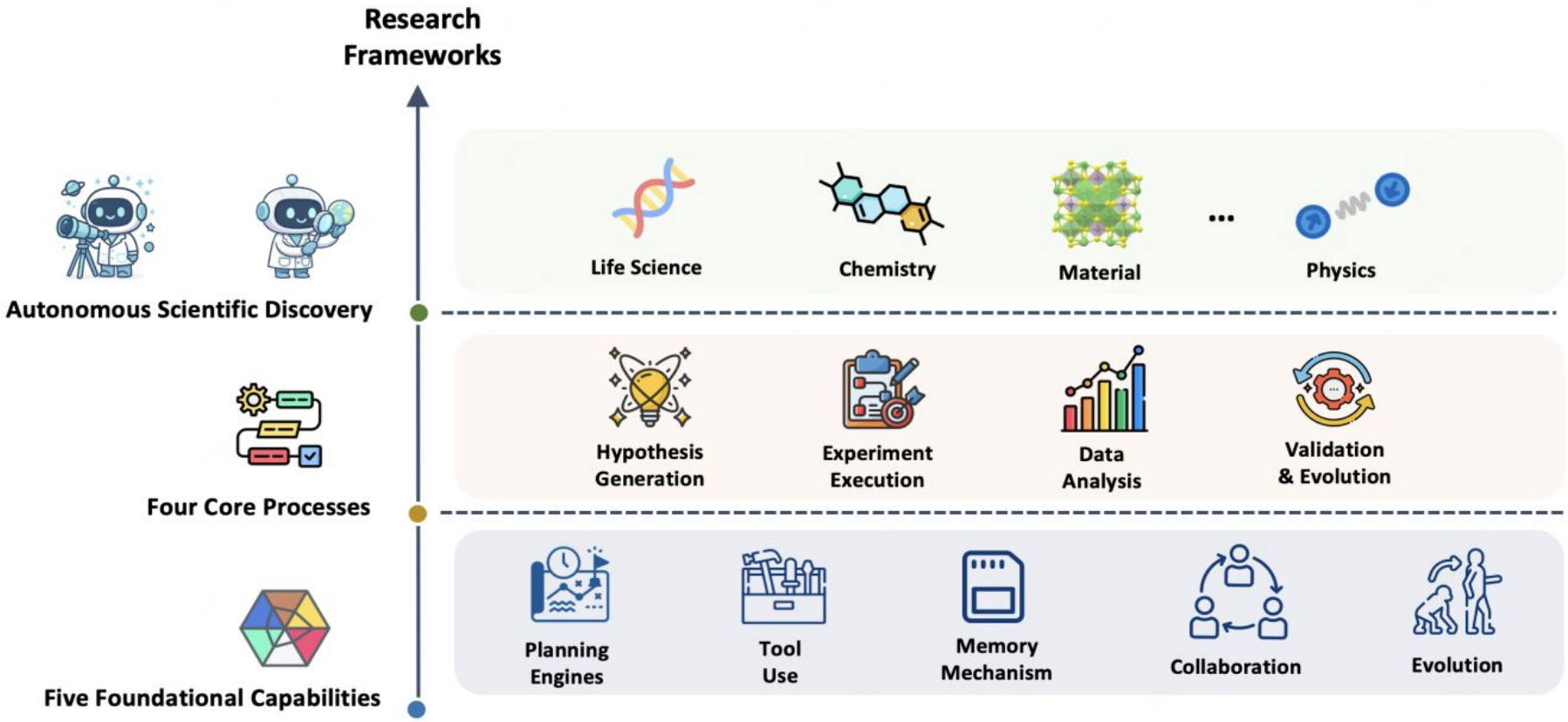


Figure 2: Research frameworks for Autonomous Scientific Discovery: Integrating Foundational Capabilities, Core Processes, and Research Levels across Life Sciences, Chemistry, Materials Science, and Physics.

Wei_2025_Agentic_Science_Survey

The Frontier of Scientific Discovery

GNoME · DeepMind, 2023

2.2 million

predicted new crystal structures



A-Lab · UC Berkeley, 2023

41 in 17 days

new materials made by a closed-loop robot lab

We will not build the robot arms. We build the brain: the workflows and loops that drive them.

Hold these numbers lightly — both results are contested. Next: both stories, twice.

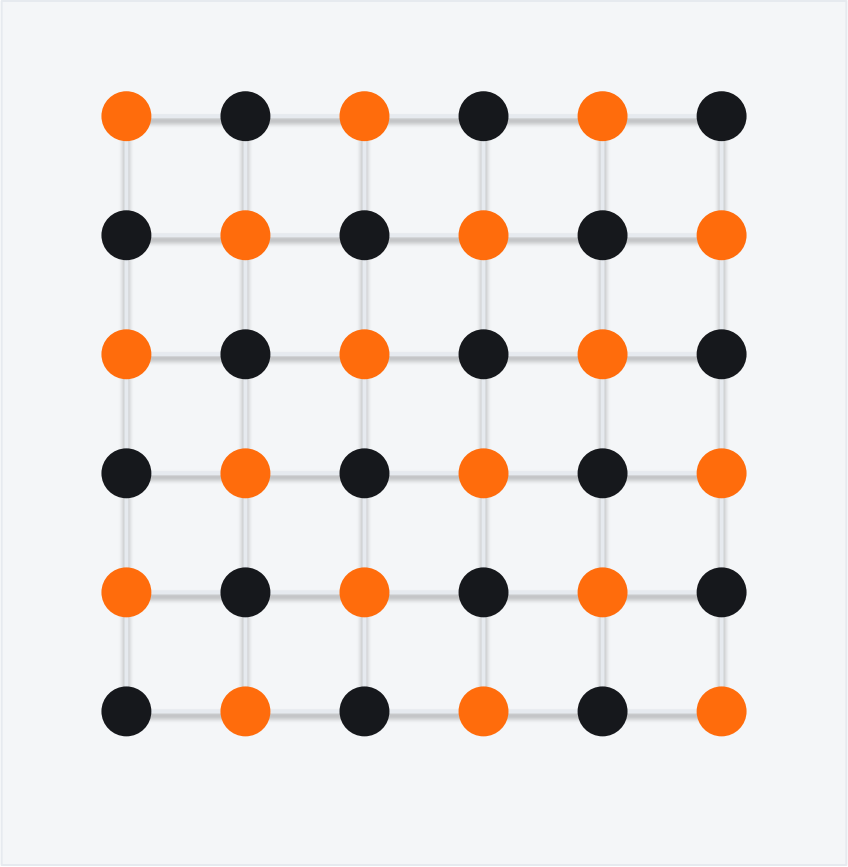
GNoME: The Headline

2.2 million

crystal structures predicted below the known stability hull — 10× the prior literature.

381,000

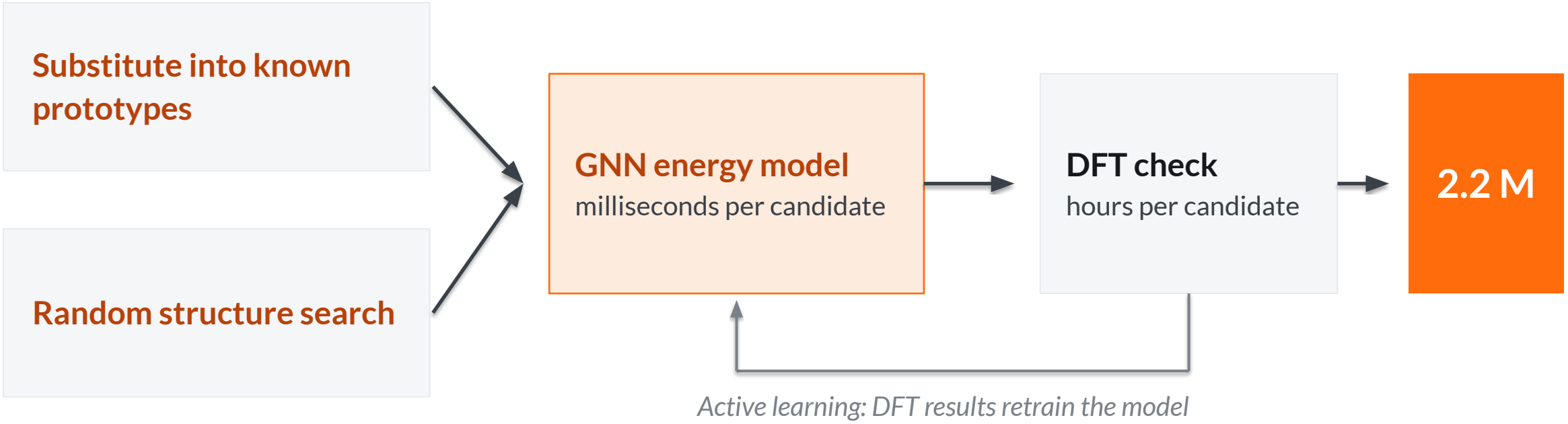
of them “newly stable and worth synthesising”.



Merchant et al., Nature 624, 80–85 (2023)

How GNoME Worked

Generate candidates two ways. Filter twice. Feed the answers back.

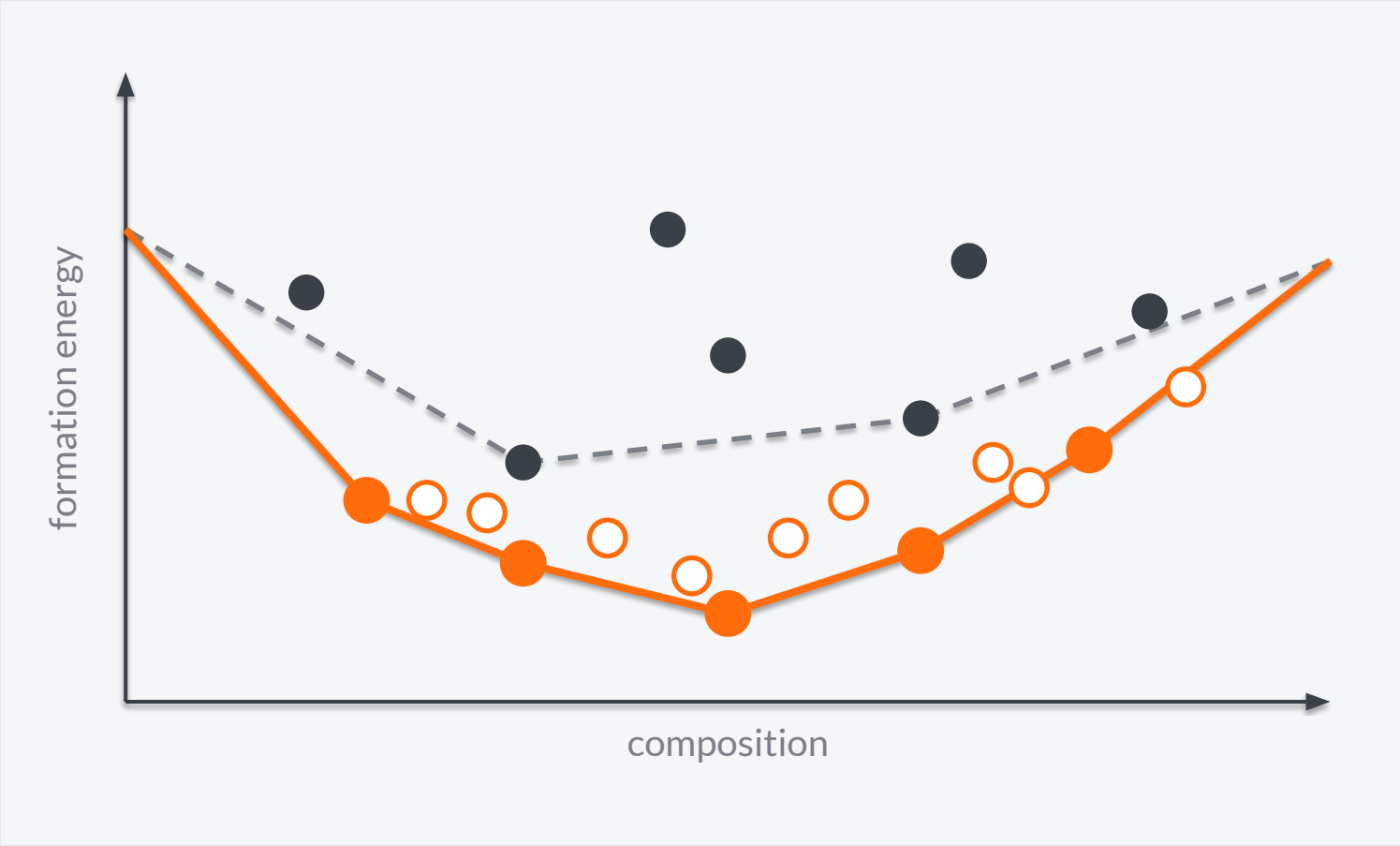


The furnace was the throughput bottleneck. Move the first screen into compute.

Merchant et al., Nature 2023

Two Numbers, Two Different Things

Both are called “stable”. Only one of them means “new”.



Known compounds

the dashed line is the old hull

2.2 million

below the old hull — but beaten by other new predictions

381,000

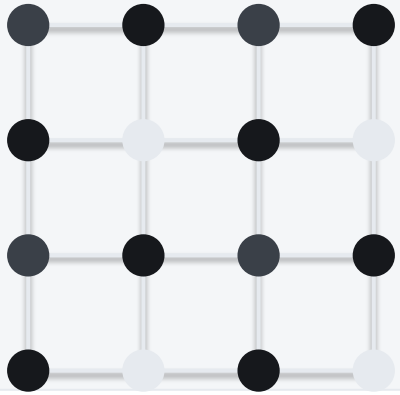
on the new hull — the honest count

Contested: New, or Just Renamed?

Cheetham and Seshadri asked a chemist's question: is any of this both new and useful?

KNOWN

ABX_3 perovskite

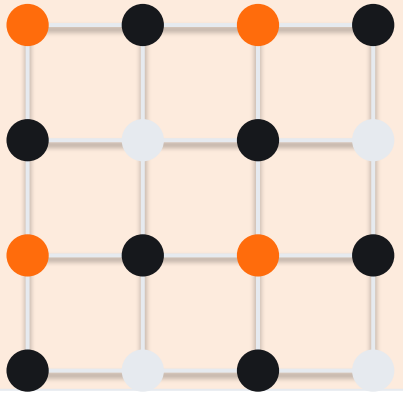


substitute



"PREDICTED"

$A'BX_3$ — one cation swapped



Scant evidence of anything both new and useful. Many were trivial substitutions. Week 8 takes this apart.

Cheetham & Seshadri, Chem. Mater. 36, 3490 (2024)

The A-Lab: Closing the Loop

The AI proposes, plans and interprets. Robots weigh, mix, bake and measure.

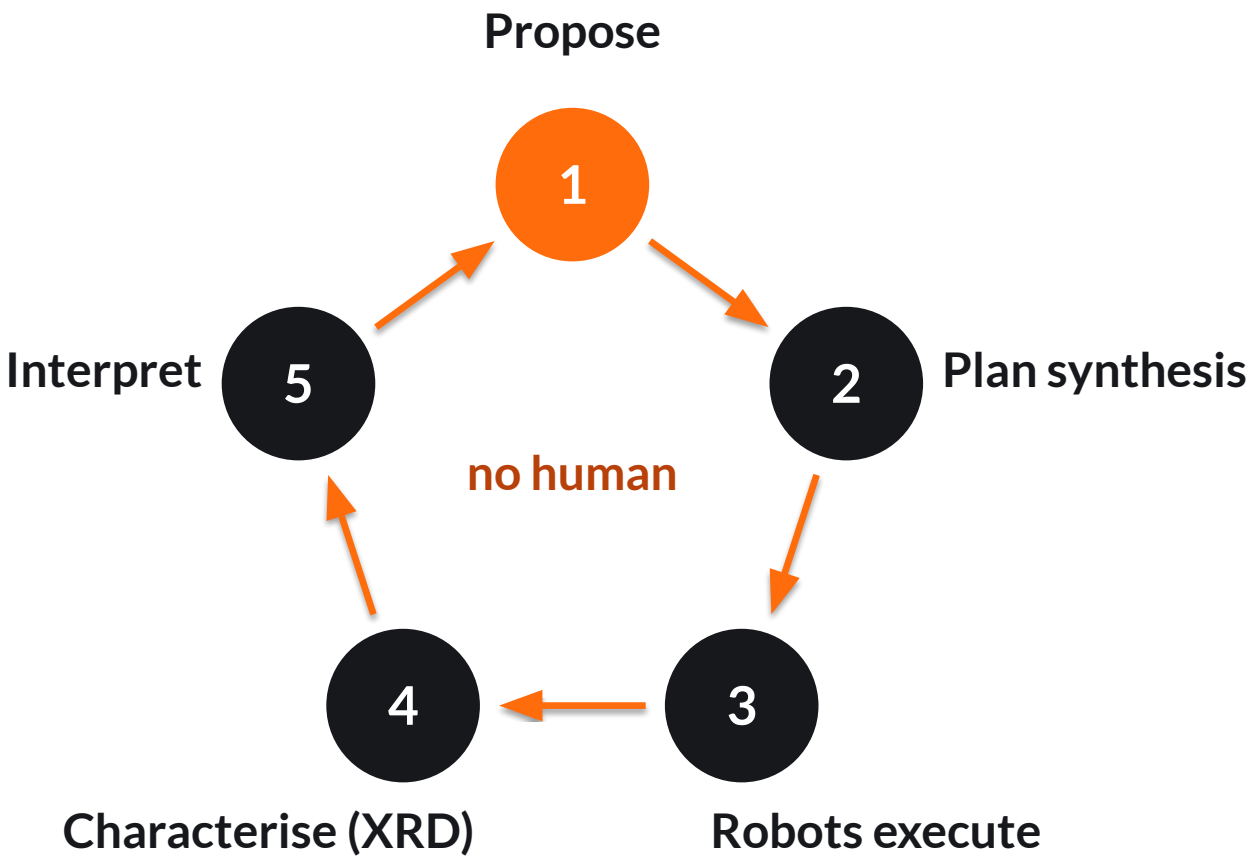




Image: newscenter.lbl.gov

Szymanski et al., Nature 624, 86–91 (2023)

Seventeen Days — and a Corrected Count

AS PUBLISHED, NOV 2023

~~41 of 58~~

“novel compounds” — 71 % success

AFTER AUTHOR CORRECTION, JAN 2026

36 of 57

targets synthesised — 63 % success

36	confirmed by hand
4	inconclusive
1	already in training data

“Novel” now means new to the platform, not new to science.

Szymanski et al., Nature 2023 · Author Correction, Nature 650, E1 (2026) · Leeman et al., PRX Energy 2024

The Lesson for You

Autonomous claims require autonomous-grade evidence.

The instrument that generates the result cannot be the only thing that validates it.

DISCUSSION · 3 MINUTES

Who validates the validator? What would convince you an autonomous lab's result is real?

BRIDGE · FROM THE OUTER LOOP TO THE SUBSTRATE

The loop fails when it trusts a raw prediction. Why does the model get the physics wrong in the first place?

PART 1
The Goal

→

PART 2
The Substrate

→

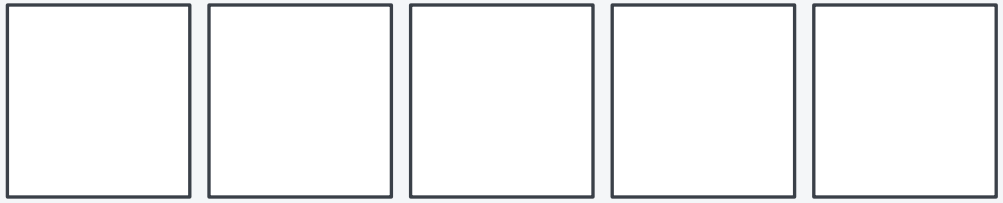
PART 3
The Interface

outer loop → physical substrate → boundary conditions

Tokenization — Discretizing Language

- An LLM does not read English. It reads numbers.
- Text is split into sub-word tokens (Byte-Pair Encoding).
- 1 token ≈ 0.75 words → 128 K tokens ≈ a 300-page book.

ENGINEERING ANALOGY — FEA MESHING



geometry → elements



text → tokens

Discretize before you can compute. A bad mesh gives artefacts; so do bad token boundaries.

Tokenization Gone Wrong

Four engineering strings, one tokenizer. Boundaries shown with |.

`350 MPa`	`|350| MP|a|`	the unit is split in two
`1.75e11`	`|1|.|75|e|11|`	nothing here looks like a number
`W12x40`	`|W|12|x|40|`	the section designation is shredded
`ε̇_pl`	`|ε|<0xCC>|<0x87>|_|pl|`	the dot falls back to raw bytes

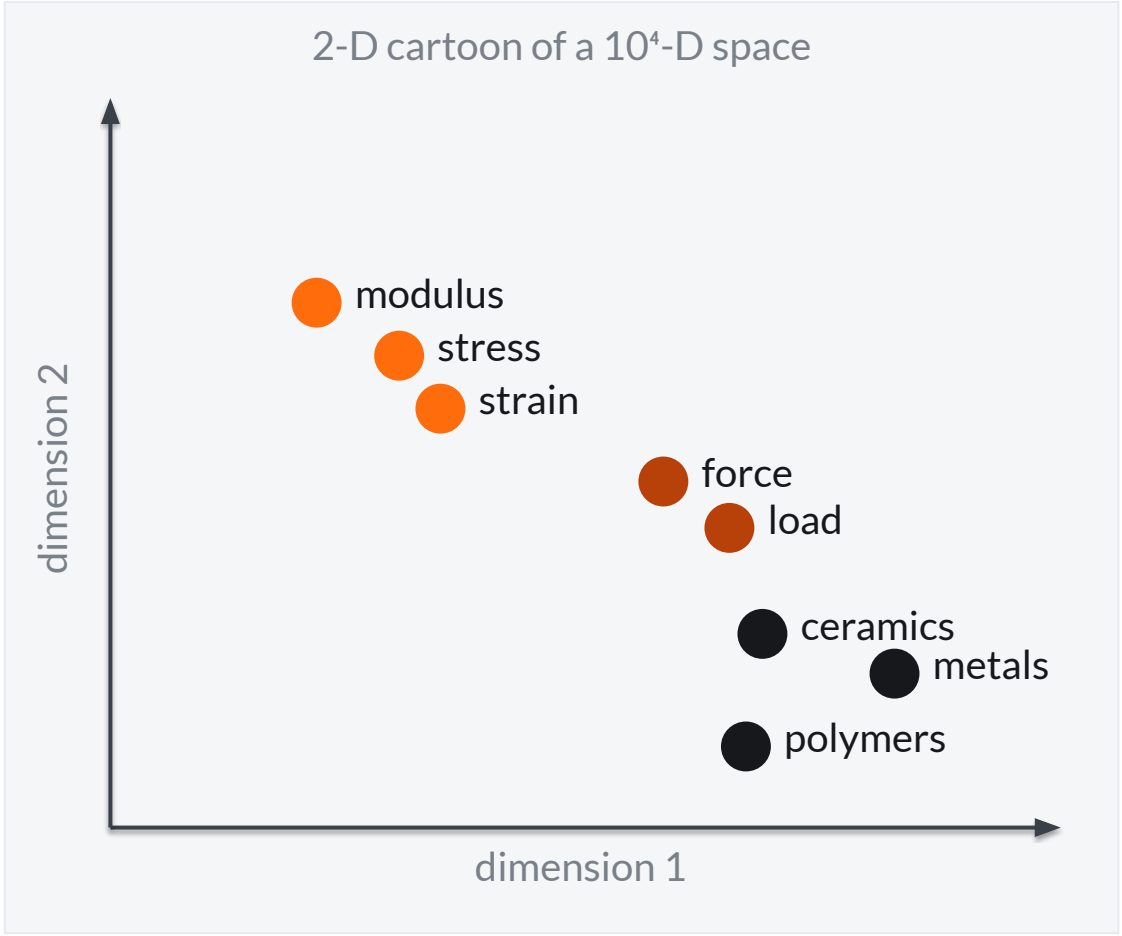
Models are bad at arithmetic and exact identifiers because of the encoding, not the reasoning.

Embeddings — Meaning as Geometry

- Each token → a vector in ~10^4 dimensions.
- Close in space ≈ similar in meaning.

ENGINEERING ANALOGY — ASHBY CHARTS

Plot materials by modulus and density: metals, ceramics and polymers cluster on their own. Plot words by learned coordinates: the mechanics vocabulary clusters on its own.



Embedding Arithmetic, on Engineering Terms

Illustrative cosine similarities. Your numbers will differ; the pattern will not.

beam deflection · girder displacement	0.81	same concept, different words — the good case
tensile strength · yield strength	0.89	two quantities you must never confuse
ASTM A992 · ASTM A36	0.94	50 ksi vs 36 ksi steel — four characters apart
stiffness · compliance	0.72	reciprocals, placed close together

Embeddings encode co-occurrence, not meaning. **In engineering that gap is dangerous.**

5-minute break

WHEN WE COME BACK

Prompts as boundary conditions — how to constrain a model that cannot multiply.

BRIDGE · FROM THE SUBSTRATE TO THE INTERFACE

The model cannot do reliable arithmetic and does not ground concepts in physics.
So we constrain it: a prompt is a boundary condition.

PART 1
The Goal

→

PART 2
The Substrate

→

PART 3
The Interface

outer loop → physical substrate → boundary conditions

Prompt Engineering: The Task

Pull material properties out of an abstract — as JSON a program can read.

THE ABSTRACT (ONE SENTENCE)

"The dissimilar friction welds, AISI 316L/IN 718 ST and AISI 316L/IN 718 STA, exhibited yield strength of 634 ± 9 MPa and 602 ± 3 MPa …"

WHAT WE WANT BACK

```
{"material": "…",
 "grade": "…",
 "yield_mpa": …}

one object per alloy
```

Two ways to ask: zero-shot (just the instruction) and few-shot (instruction plus worked examples). Same sentence, next two slides.

Beeravolu et al., Materials 16, 3049 (2023), CC BY 4.0

Zero-Shot: Ask, With No Examples

YOU SEND

```
Extract material properties
into JSON.

"The base metals were IN 718
 and AISI 316L."
```



MODEL RETURNS

```
{"base_metals":
  ["Inconel 718",
   "AISI 316L"]}
```

next run: a different key, or prose

Right content. A shape of its own — and a different one next run. Nothing is wrong. Nothing downstream can rely on it.

Few-Shot: Show the Shape You Want

YOU SEND

```
Extract material properties into JSON.

Input:  "The beam is made of A36 steel."
Output: {"material":"steel","grade":"A36"}
Input:  "We used 6061 aluminum."
Output: {"material":"aluminum","grade":"6061"}

Input:  "The base metals were IN 718
         and AISI 316L."
Output: ▌
```

REPLY A — your shape, one alloy gone

```
{"material":"steel",
 "grade":"IN 718"}
```

REPLY B — both alloys, a list nobody showed it

```
[{"grade":"IN 718", …},
 {"grade":"AISI 316L", …}]
```

Examples pin the shape — but only the shapes you showed.

Discussion: Will the Shape Generalise?

You fixed the two-alloy case with a third example. Now change the input.

THE NEW ABSTRACT

"… an FCC matrix with $L1_2$ precipitates, a σ phase at grain boundaries, and minor B2 and μ phases …"

One matrix, four phases. Your examples showed one and two.

DISCUSSION

Will the model generalise to an N-item list — or collapse back to the two it was shown?

Examples are a soft constraint. When the shape must always hold, declare it

Zero-Shot and Few-Shot: When to Pay for Examples

Zero-shot

Three runs, three shapes:
634 MPa · “approximately 634 megapascals” · YS = 634 ± 9

0

extra tokens per call. Always the right first attempt.

Few-shot

Examples fix the format — only the shapes you showed. Cover the awkward cases, including the empty one.

~360

extra tokens on every call, forever. 100 000 documents → 36 M tokens.

THE RULE Start zero-shot. Add examples when a program will parse the output. Include the empty case.

What Makes a Prompt Good (1–3)

1

State the role and the audience

Controls register and vocabulary — form, not competence.

2

Give the task, not the topic

“List the three loading conditions tested” is checkable. “Summarise” is not.

3

Specify the output format explicitly

Fields, types, units.

What Makes a Prompt Good (4–6)

4

Say what to do when the answer is not there

“If no yield strength is stated, return null.” The cheapest hallucination control you will ever write.

5

Keep the instruction out of the middle

Document first; question and format spec last.

6

Iterate against real failures, not imagined ones

Twenty inputs with known answers. Fix what broke.

Where the JSON Goes

Extraction is not the deliverable. It is stage 3 of 5.



A solver does not negotiate. A null that became 0 propagates silently into a stress result.